Sakshi Shrestha

Johnson City, Tennessee ·
https://www.linkedin.com/in/sakshi-shrestha **(423) 773-1211 ·**
sakshishrestha111@gmail.com

Education

East Tennessee State University, Johnson City, TN
Master of Science in Computer Science, May 2025
Honors: Research Assistant (Data Analysis and Visualization)

Kathmandu University, Dhulikhel, Nepal
Bachelor of Science in Computer Science, March 2019

Research Interests

Machine Learning; Artificial Intelligence; Medical Imaging; Bioinformatics; Deep Learning; Statistical Modeling; Microservices Architecture; Software Engineering; Software Security; Human-Centered and Data-Driven Computing

Publications

• Rajbhandari, Y.; Maharjan, R.; Shrestha, S.; Cerny, T. (2025). Attack Surface Score for Software Systems. Future Internet, 17(305).

• Rajbhandari, Y.; Maharjan, R.; Shrestha, S.; Cerny, T. (2025). Initial Review: Simplifying Attack Surface Scoring for Software Systems. IEEE SOSE.

• Maharjan, R.; Sooksatra, K.; Cerny, T.; Rajbhandari, Y.; Shrestha, S. (2025). Monolith to Microservices Decomposition with VAE-GNN. Future Internet, 17(303).

• Rajbhandari, Y.; Shrestha, S.; Maharjan, R.; Shaik, S. Benchmarking Motif Discovery Algorithms. ISNCC 2025.

• Shrestha, S.; Saechueng S; et al. Artificial Intelligence and Spondylolisthesis: Mechanisms of Spinal Diagnosis. Submitted to MetaRadiology (Under Review).

• Prajapati, M.; Ghimire, P.; Shrestha, S.; et al. MARVEL: Machine Learning–based Voice Evaluation Framework. Submitted to The Voice Foundation (Under Review).

Research Experience

Research Assistant – Department of Computing, East Tennessee State University
June 2025 – Current
• **AI and Spondylolisthesis**: Implemented a deep learning + ML framework for lumbar X-rays using U-Net, SMOTE, and XGBoost, achieving 96% accuracy and interpretable SHAP-based diagnostic insights.

• **Speech Pathology**: Develop a scalable ML framework for clinical voice assessment using 19,271 recordings, integrating acoustic, temporal, and longitudinal biomarkers (MFCC, spectrograms, CPPS, jitter, shimmer). Built rule-based z-score models and nonlinear RNN/LSTM classifiers for multi-pathology detection, with SHAP-based interpretability to support clinical decision-making and telehealth integration.

• **Spinal Stenosis**: Evaluate U-Net and YOLO variants on foraminal-view X-rays and engineered multi-view anatomical features for integrated ML modeling. Upcoming work includes federated learning across sites and XAI tools for interpretable stenosis diagnosis.

Research Assistant – Department of Nursing, East Tennessee State University
May 2024 – May 2025
• Conducted statistical analysis on large healthcare survey datasets using SPSS and Python
• Generated structured analytical reports.

Technical Skills

Programming: C#, Python, .NET, PHP, JavaScript, C++, C
Frameworks/Tools: Blazor, React, Angular, Laravel, WordPress, Pandas, NumPy, SPSS
Cloud & DevOps: AWS (S3, EC2, Lambda), Azure, Docker, Kubernetes, GitHub Actions, Jenkins
Databases: MySQL, PostgreSQL, MongoDB
Other: REST APIs, Data Visualization, Agile/Scrum, UI/UX Design

Professional Experience

Application Developer - ETSU, Department of Engineering, Engineering Technology, Interior Architecture, and Surveying
Oct 2025 – Present
• Building a cost-estimation software using DotNet/Blazor, PostgreSQL, Python

Blazor/Python Developer- Capstone Project – International Storytelling Center
Jan 2024 – May 2025
• Developed a microservice-based transcription pipeline using MinIO, Python, and automated workflows.
• Built and deployed a Blazor Server research platform supported by AWS-integrated object storage.
• Led a research-oriented development sub-team, coordinating sprints and maintaining documentation.

Software Developer – Vanilla Transtechnor, Kathmandu, Nepal
Aug 2021 – Apr 2023
• Developed scalable WordPress platforms for enterprise clients.
• Built LMS infrastructure based on Moodle supporting educational workflows.
• Supported testing and QA processes for remittance software systems.

Junior Software Developer – Eight Square Pvt. Ltd., Nepal
Dec 2018 – Aug 2021
• Designed responsive and accessible web interfaces using HTML5, CSS, and SASS.
• Developed backend features for an elder-friendly fitness application using Laravel.

Freelance Developer – Kathmandu, Nepal
Mar 2019 – Aug 2021
• Designed and built the COVID-19 Nepal Information Portal.
• Produced cross-media UI/UX assets for award-winning MIT/GSL projects in 2019.